Ansh Chandrakar

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ABOUT ME

I'm a curious and motivated physics student with a strong interest in science and technology. I love exploring new ideas and enjoy learning both in and outside the classroom. I've always been fascinated by astronomy and astrophysics, and I'm excited to keep learning more about these areas. Lately, I've also been really interested in quantum computing and its potential. I'm always looking for ways to grow—through projects, research, or hands-on experience—and hope to contribute to the fields I'm passionate about.

EDUCATION

Master of Science in Physics 2025 - Present
Indian Institute of Technology, Hyderabad, Telangana

- *Core Courses*: Electrodynamics, Classical Mechanics, Quantum Mechanics - 1, Statistical Mechanics, Mathematical Physics - 1

Bachelor of Science (Honours) in Physics 2022 - 2025
Birla Institute of Technology, Mesra, Ranchi, Jharkhand

- *CGPA*: 8.6/10
- *Core Courses*: Mechanics, Mathematical Physics - 1, 2 and 3, Waves and Optics, Electricity and Magnetism, Thermal Physics, Digital Systems and Applications, Elements of Modern Physics, Analog Systems and Applications, Quantum Mechanics and Applications, Solid State Physics, Electromagnetic Theory, Statistical Mechanics
- *Elective Courses*: Computational Physics Skills, Radiation Safety, Nuclear and Particle Physics, Classical Dynamics, Astronomy and Astrophysics

High School 2017 - 2021
Sri Sankara Vidyalaya, Bhilai, Chhattisgarh

- *12th*: 91.8%
- *10th*: 92.4%
- *Major Courses*: Physics, Chemistry, Mathematics

RESEARCH PROJECTS

MSc Winter Project: Non-Reciprocal Ising Model with Stochastic Resetting

- Studied non-equilibrium statistical mechanics of Ising systems with asymmetric (non-reciprocal) interactions that violate detailed balance.
- Analyzed stochastic spin-flip dynamics and the emergence of non-equilibrium steady states with probability currents.
- Investigated the effects of stochastic resetting on system dynamics and steady states, highlighting the interplay between non-reciprocity and resetting.
- Explored broader implications for non-equilibrium phase behavior, with extensions proposed to Potts and XY-type models.

Pedagogical Study on Quantum Entanglement (Manuscript in Preparation)

- Preparing a pedagogical article focused on understanding quantum entanglement through the lens of coupled quantum harmonic oscillators, aiming to make the concepts accessible to a wider physics audience.

- Developing clear demonstrations and illustrations using Python (QuTiP) simulations to highlight key ideas such as Von Neumann entropy, entanglement dynamics, and decoherence.
- Targeting submission to *Resonance* journal to share findings and teaching-oriented insights.

Bachelor's Dissertation: Entanglement in Coupled Quantum Oscillators Using QuTiP

- Explored entanglement in coupled quantum harmonic oscillators using the Quantum Toolbox in Python (QuTiP) as part of the Bachelor's dissertation project.
- Investigated coupling mechanisms, unitary evolution, and computed entanglement measures such as Von Neumann entropy and logarithmic negativity across various systems.
- Extended the study to hybrid systems coupling quantum oscillators with coherent states, analyzing entanglement dynamics and decoherence effects.

SKILLS

Computational

- MATLAB
- Python (QuTiP, NumPy, SciPy)
- Fortran
- Linux (Command line, Bash scripting, System administration basics)

Typesetting

- LaTeX

Languages

- Hindi: Mother tongue
- English: Fluent

EXPERIENCE

Winter School on Recent Frontiers in Physics Research (Online) December 2025

Indian Institute of Science (IISc), Bengaluru

- Participated in advanced lectures covering astrophysics and cosmology, quantum condensed matter, soft and active matter, plasma physics, photonics, and quantum technologies.
- Took part in tutorial sessions involving computational demonstrations and numerical exercises to illustrate physical concepts.
- Engaged with faculty and researchers through interactive discussions, with particular interest in lectures related to astrophysics and cosmology.
- Certificate of Participation

IIA Online Summer School June 2025

Indian Institute of Astrophysics, Bengaluru

- Selected to participate in a summer school hosted by the Indian Institute of Astrophysics.
- Attended lectures and hands-on sessions on advanced topics including galactic structure, observational techniques, radio astronomy, cosmology, and solar physics.
- Engaged with leading scientists through interactive sessions to deepen conceptual and practical understanding of astrophysics.
- Certificate of Completion

Society for Data Science (Member) Feb 2023 – May, 2025

BIT, Mesra, Ranchi

- Participated in Python workshops to enhance programming skills.
- Worked in organizing society events and managing logistics.
- Collaborated with the team to gather sponsors for events.

CERTIFICATIONS & AWARDS

- BIT GP Birla Scholarship for Program Topper – Awarded twice (2 semesters) for outstanding academic performance at BIT Mesra